

Razia Sultana Jui

Research Interests: Security and Privacy + (1)-682-227-1680 | raziarazzak1@gmail.com

EDUCATION

BRAC University

Bachelor of Science in Computer Science and Engineering
Department of Computer Science and Engineering

Summer, 2018 - Fall, 2023

CGPA (Majors) : 3.25/4.00

RESEARCH PROJECT EXPERIENCE

Identifying Hate Speech of Bangla Language Text Using Natural Language Processing | [Link](#)
Natural Language Processing, Deep Learning, Hate Speech Detection, Bangla Text Classification **2024 – 2025**

- Pursued undergraduate thesis under the supervision of **Mr. Annajiat Alim Rasel**, Senior Lecturer, Department of Computer Science and Engineering, BRAC University.
- Developed deep learning models leveraging **BiLSTM**, **CNN-LSTM**, and transformer-based architectures (**mBERT**, **XLM-R**) to detect hate speech in Bangla text.
- Designed a comprehensive preprocessing pipeline to normalize and clean Bangla language data, improving classification performance and robustness.
- Contributed to enhancing online content moderation and promoting safer digital communication within Bangla-speaking communities.

PROFESSIONAL PROJECTS

Gift Cards Zone – E-commerce Website Development | *WordPress, PHP, CSS* June 2021 | [Website](#)

- Built a full-stack e-commerce platform for a multi-purpose retail brand using WordPress and PHP.
- Implemented subscription-based reviews, sales tracking, and admin recommendation features to enhance engagement and management efficiency.

ACADEMIC PROJECTS

Fire and Smoke Detection System using Arduino | *C++, Arduino Uno, Sensors (DHT22, MQ2)* Summer 2022

- Designed an automated safety system to detect fire, smoke, and gas leaks using flame and MQ2 gas sensors to mitigate industrial risks.
- Integrated DHT22 sensors for real-time temperature and humidity monitoring, displaying live data on an LCD interface.
- Implemented an immediate alert mechanism using buzzers and LED indicators to signal hazard detection effectively.

Heart Disease Prediction using Machine Learning | *Python, Scikit-learn, Pandas, NumPy* Fall 2022

- Developed a diagnostic model using a Kaggle dataset of 1026 entries, implementing KNN, Decision Tree, and Logistic Regression algorithms.
- Achieved 93% accuracy by applying MinMax and Standard Scaler normalization techniques to optimize model performance.

- Evaluated classification reliability using confusion matrices, precision, recall, and F1-scores to ensure robust disease detection.

Real Estate Management System | *Django, Python, HTML, CSS, JavaScript*

Spring 2021

- Built a full-stack web platform facilitating property trading, featuring a dynamic 48-hour bidding system to optimize sales value.
- Engineered role-based access for admins and users, enabling secure property listings, price filtering, and user verification.
- Designed the system to eliminate intermediaries, effectively reducing transaction costs by 9% for rural property buyers.

HIGHLIGHTED ACADEMIC COURSES

- | | |
|------------------------------------|---------------------------------------|
| • CSE-423 Computer Graphics | • CSE-420 Compiler Design |
| • CSE-425 Neural Networks | • CSE-360 Computer Interfacing |

TECHNICAL SKILLS

Languages: Java, C, C++, Python, Matlab, JavaFX, Gnu-Plot, Bash, Assembly, Swing

Database: Oracle, MySQL, SQLite, MongoDB

OS: Windows, Linux (Kali, Ubuntu)

Version Control: Git (GitHub)

Frameworks: JavaFX, Django, Node.js, React

Web Technology: HTML, CSS, PHP, Bootstrap, WordPress, JSON, XML

Compiler Tools: Flex, Bison, EMO 8086

Micro-Controller: Arduino, ATmega 32, Atmel Studio, Proteus 832

Technical Writing: $\text{\LaTeX}$, Beamer, Overleaf

Other: Networking Simulator 3 (NS3), iGraphics, Cisco-Packet-Tracer, PyTorch, OpenGL

REFERENCES

Dr. Muhammad Iqbal Hossain

Assistant Professor, Department of Computer Science and Engineering
BRAC University, Dhaka, Bangladesh.

Email:

iqbal.hossain@bracu.ac.bd

Relation: Thesis Supervisor

Dr. Muhammad Iqbal Hossain

Assistant Professor, Department of Computer Science and Engineering
BRAC University, Dhaka, Bangladesh.

Email:

iqbal.hossain@bracu.ac.bd

Relation: Thesis Supervisor

Mr. Annajiat Alim Rasel

Assistant Professor, Department of Computer Science and Engineering
BRAC University, Dhaka, Bangladesh.

Email: annajiat@bracu.ac.bd

Relation: Research Supervisor